Columbia DBMI Summer Research Program 2025

PROGRAM DEADLINE : February 14, 2025 at 11:59 PM(Midnight)

Applicant Information

Name:	Selina Hui
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App ID:	05095483
Status:	Submitted
Cycle:	None
Last Modified:	Feb 14 2025 23:31 EST by selina.hui2005@gmail.com
Last IP Address:	34.120.172.148

Applicant Information

Personal Information

First name	Selina Wai Yan
Middle name	
Last name	Hui
Date of Birth **Note: applicants must be at least 16 years old.**	01/06/2005
Rising/graduating seniors in high school and college or university undergraduate students who will be in the New York City area during the summer of 2025 are invited to apply for the Columbia DBMI Summer Research Program. As noted above, applicants must be at least 16 years old. **Please list your citizenship status (US citizen, US Permanent Resident) or current VISA status if not a citizen or permanent resident.**	US Citizen

Personal Contact Information

Street Address	13934 Bergenfield Dr
City	North Potomac
State	MD
Zip or Postal Code	20878
Phone Number	213-691-3823
Email	selina.hui2005@gmail.com

Current School Enrollment

Please list the name of the school (high school or college) in which you are currently enrolled.	University of Southern California
City in which school is located	Los Angeles
State in which school is located	CA

Program Referral

How did you hear about the program?	Friend or colleague~Internet search (please specify search below)~LinkedIn

Please described exactly how you first heard about the program if not specified above

Online search for summer research programs in the field of biomedical informatics

Personal Statement

In a personal statement (no more than 300 words) please answer the following question:

What are your goals in participating in the DBMI Summer Research Program?

My primary goal for participating in the DBMI Summer Research Program is to gain experience in integrating advanced computational methods, mobile technologies, and large-scale data analysis to address critical challenges in neuroscience research and improve patient outcomes. I aim to develop expertise in handling and analyzing neurological disease datasets from clinical records, applying AI and machine learning techniques to extract meaningful insights. I am eager to enhance my skills in machine learning models and data assimilation, particularly for precision medicine and disease phenotyping. As I plan to pursue a Ph.D in bioinformatics, this program offers a unique opportunity to immerse myself in the multidisciplinary environment of DBMI, bridging my background in computational neuroscience with broader applications in biomedical informatics. These skills would allow me to make meaningful contributions to the field by advancing our understanding of brain function and improving human health. I'm particularly excited about the possibility of creating predictive models for stroke outcomes using large-scale data sources and developing digital twins for precision medicine in neurodegenerative diseases like Alzheimer's, integrating multi-omics data, clinical information, and environmental factors. Additionally, this program will improve my ability to design and present research through participation in lab meetings and journal clubs, honing my critical thinking and scientific communication skills. I am excited to collaborate with peers and mentors in the DBMI Summer Research Program, contributing to innovative research while building a network to support my future in biomedical informatics.

Resume/CV (curriculum vitae)

Upload Resume

Please attach your most recent resume/cv (curriculum vitae)	Hui, Selina - Resume (R).pdf

Programming Skills

Programming Skills

Please list your programming skills (linking to a code repository if you have any is welcome) as well as any research or professional experience you may have had in information technology or data science that you think is relevant.

Programming Skills:

- Python
- C++, C
- Machine Learning related toolkits including TensorFlow, Keras, Pandas, NumPy, SciPy, matplotlib, scikit-learn
- JavaScript, Bash, HTML (for web-based experiments), MATLAB
- MySQL and MySQL Workbench, NoSQL, MongoDB: Currently learning through college coursework
- Genomic analysis tools and databases: NCBI BLAST (Basic Local Alignment Search Tool), CARD (Comprehensive Antibiotic Resistance Database), UniProt, KEGG, GenBank

Research Experience:

1. USC Computational Neurorehabilitation Laboratory (Current):

- Designing and implementing online motor learning experiments using Python, JavaScript, HTML, and Bash
- Deploying experiments via Prolific and Firebase CLI
- Analyzing visuomotor adaptation in stroke patients

2. Neurotech@USC Project Lead (Current):

- Investigating bilingual code-switching using eye-tracking data
- Applying machine learning to analyze neural mechanisms of language processing

3. UniversumX EEG Data Analysis (Current):

- Applied machine learning techniques to analyze complex EEG sensory data

4. Artificial Intelligence Applications Coursework (Fall 2024):

- Implemented DNNs, CNNs, and RNNs for biomedical applications
- Designed a CNN for multi-class image classification, demonstrating skills in neural architecture creation, data preprocessing, and model optimization

5. High School Research - Molecular Biology Laboratory (2019-2023):

Studied antibiotic resistance in horse gut E. coli

Combined wet-lab techniques with bioinformatics and genomic sequencing using BLAST, CARD

Published three genomic sequences in NIH NCBI GenBank (Accession Numbers: CP091839, NZ_CP091773, NZ_CP091836)

Recommendation Letters

Recommendation Letter

Contact's First Name	Dion
Contact's Last Name	Dickman
Contact's Email	dickman@usc.edu
Contact's Phone Number	
Contact's Title	Professor
Contact's Organization	University of Southern California

Recommendation Letter

Contact's First Name	Nicolas
Contact's Last Name	Schweighofer
Contact's Email	schweigh@pt.usc.edu
Contact's Phone Number	323-442-1838
Contact's Title	Dr.
Contact's Organization	University of Southern California

Transcripts

Instructions

Transcripts

You may upload a transcript here.	Hui_Selina_Unofficial_School_Transcript.pdf
You may upload a transcript here.	
You may upload a transcript here.	

Additional Applicant Info

Dismissal or suspension

Have you ever been dismissed or suspended from an academic institution? If yes, please explain fully below.	No
Explanation, if you answered "yes":	

Submit Application

By entering your full name below, you certify all answers are truthful and accurate.	Selina Hui
If you are under 18, we require that your parent or guardian enter their full name to certify that all answers given in this application are truthful and accurate and that they consent to you applying to this program.	